Market Risk of a Commodity Inventory: VaR and Expected Shortfall Four Ways

Juan Carlos Chavarría Morales
M.Sc. Optimization, Hochschule Bielefeld (HSBI)
Module *Risikomanagement*, winter term 2024/25

October 2026

Abstract

A German company holds 500 tonnes of silver. Silver is priced in US dollars, the company reports in euros: the euro value of the stock moves with the silver price and with the exchange rate. We measure the one-day 95% Value at Risk (VaR) of this exposure from one year of daily data (November 2022 to November 2023), extend it to a book of silver and copper, and compare five methods: parametric (delta-normal), historical simulation, Monte Carlo with normal and Student-t shocks, and filtered historical simulation with EWMA volatility. The VaR of the silver stock (EUR 334 m) is EUR 8.3 m, i.e. 2.5 % of its value; ignoring the correlation between silver and the dollar would overstate it by 18 %, because the two act as a natural hedge in euros. For the two-commodity book the one-day estimates of the methods differ by at most 18 % and differ most in the tail (99% ES, ten days). All methods pass an out-of-sample backtest on the course data, but half a year of backtest has little power: a 99% model with twice the nominal exception rate is rejected only one time in eight. On eleven years of public data (2016–2026) the backtest separates the methods: the VaRs with normal tails are exceeded almost twice as often as they should be at 99%, and only historical and filtered historical simulation pass, the latter with the best coverage.

1 The exposure

Let q_i be the quantity of commodity i (in the unit of its price), S_i its price in USD and x the EUR/USD rate (USD per EUR). The euro value of the book is

$$V = \sum_i V_i, \qquad V_i = \frac{q_i S_i}{x}. \tag{1}$$

With the risk factors $s_i = \log S_i$ and $f = \log x$, a scenario of factor changes $(\Delta s, \Delta f)$ revalues the book exactly (*full revaluation*) as

$$\Delta V = \sum_i V_i \left(e^{\Delta s_i - \Delta f} - 1 \right) \approx \sum_i V_i (\Delta s_i - \Delta f), \tag{2}$$

the right-hand side being the delta approximation. A rise of the dollar ($\Delta f < 0$) raises the euro value; when silver and the euro rise together, the two effects cancel. The loss is $L = -\Delta V$; for a level α, VaR_α is the α-quantile of L and the expected shortfall $\mathrm{ES}_\alpha = \mathbb{E}[L \mid L \geq \mathrm{VaR}_\alpha]$ the mean loss beyond it (McNeil et al., 2015). ES is a coherent risk measure (Artzner et al., 1999); VaR is not, but remains the reference in reporting.

The case. The course exercise gives 500 t of silver (16.1×10^6 troy ounces) and asks for the one-day VaR(95%) of its euro value, "taking into account the correlation that follows from the data". As a second position we add 40 000 t of copper; the course gives no quantity, so it is chosen to make the two positions of similar size (EUR 334 m and EUR 300 m on 10 November 2023).

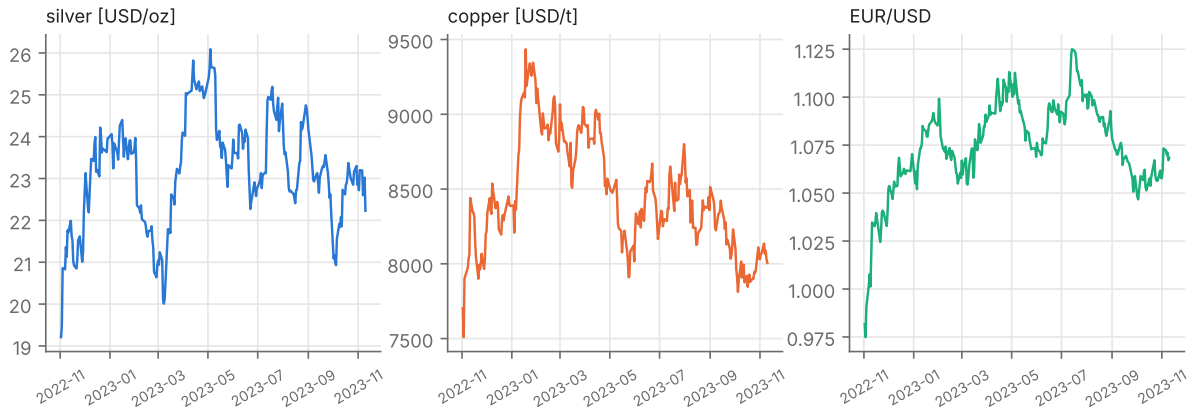


Figure 1: Daily closes used in the study (weekdays common to the three series).

Data. Daily closes of silver (USD/oz), copper (USD/t) and EUR/USD from 2 November 2022 to 10 November 2023, as provided in the course (source: finanzen.net). Silver and EUR/USD also contain weekend quotes; on the 258 weekdays on which all three closed we have 257 daily returns (Figure 1). Annualised volatilities are 27 % (silver), 20 % (copper) and 9 % (EUR/USD); silver correlates with EUR/USD at +0.48 and with copper at +0.21. All three return series have excess kurtosis (2.2, 1.4 and 0.7): fat tails, one of the stylised facts of financial returns. The course data are not redistributed; the repository contains a script that builds the same table from public sources (Chavarría Morales, 2026).

2 Methods

Parametric (delta-normal). Factor changes normal with the sample covariance Σ, delta approximation: $L \sim N(0, \delta^\top \Sigma \delta)$ with $\delta = (V_1, \ldots, V_n, -V)$, so $\mathrm{VaR}_\alpha = z_\alpha \sigma_L$ and $\mathrm{ES}_\alpha = \sigma_L \varphi(z_\alpha)/(1-\alpha)$.

Historical simulation. Today's book revalued under every observed (overlapping) h-day factor change; no distributional assumption, but only as many scenarios as days.

Monte Carlo. 200 000 correlated scenarios (Cholesky factor of Σ) with full revaluation, with normal shocks or multivariate Student-t shocks of the same covariance; the degrees of freedom ($\nu = 8.2$) are fitted to the daily P&L of the book.

Filtered historical simulation. Each factor's returns are divided by their EWMA volatility ($\lambda = 0.94$, J.P. Morgan/Reuters, 1996) and rescaled by today's forecast (Hull and White, 1998): the empirical joint shape of the shocks is kept, volatility clustering is taken into account.

For ten days, the parametric and Monte Carlo methods scale the covariance by 10 (square-root-of-time); the historical methods use overlapping ten-day returns. Risk contributions follow the Euler allocation of the parametric VaR, $C_i = z_\alpha \, \mathrm{cov}(L_i, L)/\sigma_L$, which adds up to the VaR of the book (Tasche, 1999).

3 The course exercise: 500 t of silver

The one-day VaR(95%) of the silver stock is **EUR 8.28 m**, 2.5 % of its value of EUR 334 m; the ES(95%) is EUR 10.4 m. The exercise asked for the correlation, and it matters: with the same volatilities and zero correlation, the VaR would be EUR 9.76 m, 18 % higher. Silver and the euro

Table 1: VaR and ES of the silver and copper book (EUR m), as of 10 November 2023.

	95%				99%			
	1 day		10 days		1 day		10 days	
method	VaR	ES	VaR	ES	VaR	ES	VaR	ES
parametric (delta-normal)	10.4	13.1	33.0	41.4	14.7	16.9	46.6	53.4
historical simulation	9.8	12.4	30.2	35.0	14.7	16.2	39.4	41.2
Monte Carlo, normal	10.3	12.9	31.8	39.7	14.6	16.7	44.7	51.0
Monte Carlo, Student t	10.1	13.6	31.1	41.6	15.7	19.4	47.9	58.8
filtered HS (EWMA)	8.8	13.1	27.4	39.7	14.7	19.0	45.4	57.4

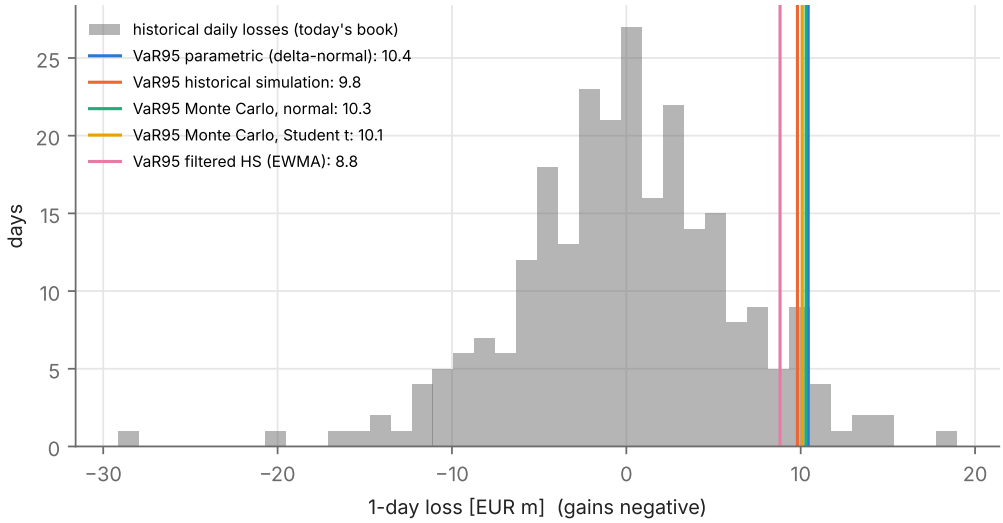


Figure 2: One-day losses of today's book under the 257 historical scenarios, and the one-day VaR(95%) of the five methods.

tend to rise together against the dollar ($\rho = +0.48$), so part of a fall of silver in USD is offset in EUR by a weaker euro and vice versa: holding silver in a euro balance sheet is less risky than its dollar volatility suggests (silver price risk alone: EUR 9.29 m).

4 The book: silver and copper

Table 1 compares the methods. At the one-day horizon and 95%, they differ by at most 18 % (EUR 8.8–10.4 m); the filtered historical simulation is lowest because volatility at the end of the sample was below its one-year average. At 99% and ten days the differences grow: the Student-t Monte Carlo and the filtered historical simulation have the largest ES (EUR 57–59 m), the plain historical simulation the smallest, because 248 overlapping ten-day returns contain few independent extreme scenarios. Figure 2 shows the historical loss distribution of today's book with the five one-day VaR estimates.

Diversification. The stand-alone VaRs of silver and copper add up to EUR 14.2 m; the VaR of the book is EUR 10.4 m, 27 % less (Table 2 and Figure 3). Silver contributes two thirds of the book VaR for a little over half of its value: copper is less volatile and only weakly correlated with silver.

Table 3: Out-of-sample backtest of the one-day VaR (rolling window of 125 days, 132 test days).

method	level	expected	exceptions	Kupiec p	Christoffersen p
parametric (delta-normal)	95%	6.6	5	0.51	0.62
historical simulation	95%	6.6	7	0.87	0.45
Monte Carlo, normal	95%	6.6	5	0.51	0.62
filtered HS (EWMA)	95%	6.6	7	0.87	0.45
parametric (delta-normal)	99%	1.3	2	0.58	0.80
historical simulation	99%	1.3	3	0.21	0.76
Monte Carlo, normal	99%	1.3	2	0.58	0.80
filtered HS (EWMA)	99%	1.3	3	0.21	0.76

Table 2: Euler allocation of the one-day parametric VaR(95%) (EUR m).

position	value [EUR m]	stand-alone VaR	contribution
silver	334	8.28	6.82
copper	300	5.92	3.61
book	634	14.20	10.43

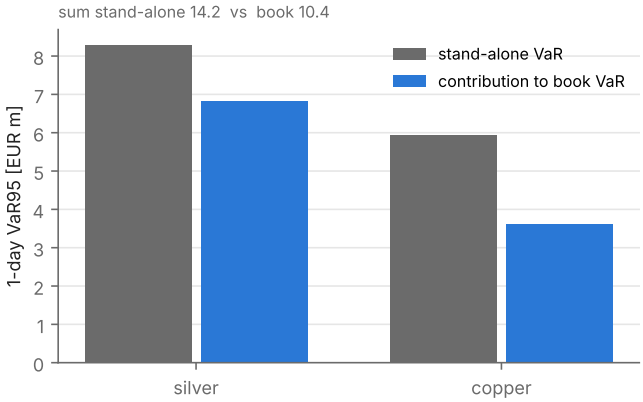


Figure 3: Stand-alone VaR and contribution to the book VaR of each position.

5 Backtest

Each day from May 2023, the one-day VaR is estimated from the previous 125 days and compared with the realised loss of the book on the next day (132 days, Figure 4). The Kupiec test checks the number of exceptions (Kupiec, 1995), the Christoffersen test their independence (Christoffersen, 1998). All methods pass both at 95% and 99% (Table 3).

A passed backtest of this length is weak evidence. Table 4 gives the probability that the Kupiec test rejects a model whose true exception rate is too high: a 95% VaR that is exceeded on 10 % of the days is caught in about half of the cases, a 99% VaR exceeded on 2 % of the days in one case out of eight. Regulatory backtests use 250 days or more for this reason, and at the 99% level even a year leaves little power.

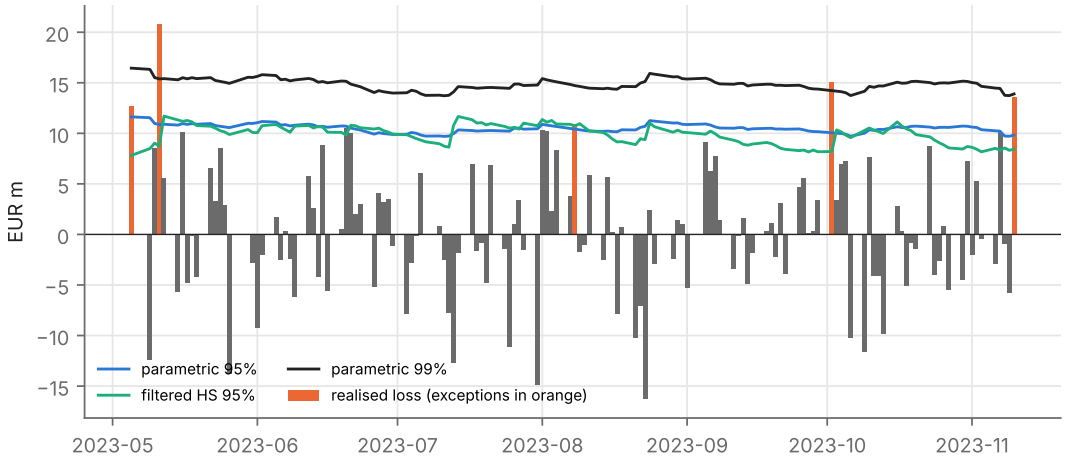


Figure 4: Realised daily losses of the book (exceptions of the parametric 95% VaR in orange) and the rolling VaR forecasts.

Table 4: Power of the Kupiec test with 132 backtest days (5% significance).

VaR level	true exception rate	probability of rejection
95%	7.5%	19%
95%	10.0%	57%
95%	15.0%	97%
99%	1.5%	5%
99%	2.0%	13%
99%	3.0%	36%

6 Backtest on eleven years of public data

The course data cover one year. With `scripts/fetch_data.py` the same book (the quantities of Section 4 held constant) was backtested on public data: the ECB reference rate and front-month COMEX futures for silver and copper (Yahoo Finance), 2 January 2015 to 16 October 2026. The VaR is estimated each day from the previous 250 days, giving 2686 test days from January 2016 on. The prices are not redistributed; the repository keeps the test statistics (`results/public_snapshot`).

Over this period the returns are far from normal: the excess kurtosis of the daily log returns is 36 for silver and 27 for copper, and a multivariate Student-t fit gives $\nu = 3.5$ degrees of freedom (8.2 on the course year). The largest moves are single days: silver fell by 31 % on 30 January 2026, copper by 22 % on 31 July 2025, when the US exempted refined copper from its import tariff. Front-month futures also contain roll effects; both make the tails heavier than in a spot series.

Now the backtest separates the methods (Table 5 and Figure 5):

- *Normal tails fail at 99%.* The delta-normal and the normal Monte Carlo VaR are exceeded 45 and 48 times instead of 27, Kupiec $p = 0.001$ and 0.0002. At 95% their counts are right (122 and 127 against 134): the normal distribution is adequate in the body of the distribution and wrong in the tail.

- *Constant volatility gives clustered exceptions.* At 95% the delta-normal VaR has the right number of exceptions, but they come in runs (Christoffersen $p = 0.03$): after March 2020 the 250-day volatility adapts too slowly, and the exceptions of a whole month count against the model.

Table 5: Out-of-sample backtest on public data (rolling window of 250 days, 2686 test days, 2016–2026).

method	level	expected	exceptions	Kupiec p	Christoffersen p
parametric (delta-normal)	95%	134.3	122	0.269	0.030
historical simulation	95%	134.3	147	0.268	0.294
Monte Carlo, normal	95%	134.3	127	0.514	0.053
filtered HS (EWMA)	95%	134.3	135	0.951	0.932
parametric (delta-normal)	99%	26.9	45	0.001	0.045
historical simulation	99%	26.9	35	0.132	0.085
Monte Carlo, normal	99%	26.9	48	0.000	0.064
filtered HS (EWMA)	99%	26.9	33	0.250	0.067

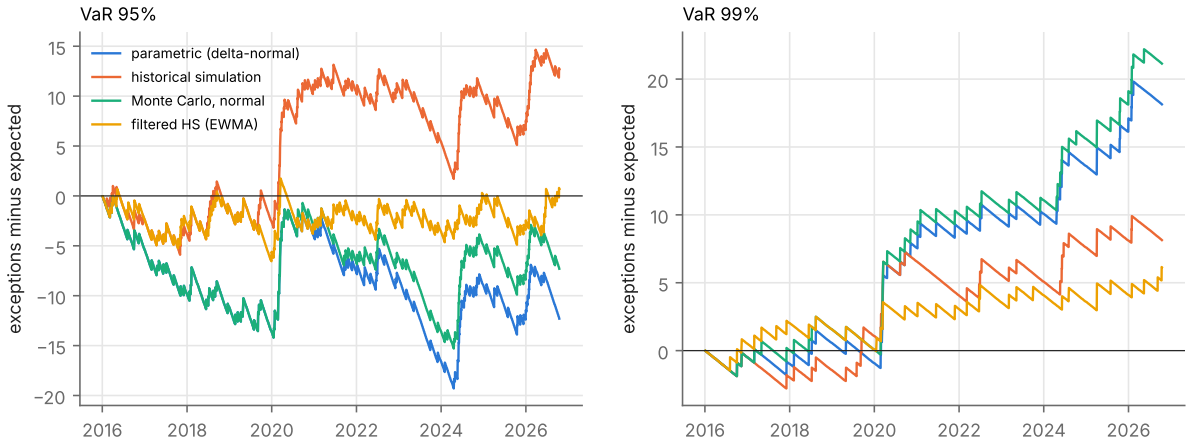


Figure 5: Cumulative number of VaR exceptions minus the expected number, 2016–2026. A correct VaR wanders around zero; a VaR that is too low drifts upwards; clustered exceptions appear as steps (March 2020, the copper and silver crashes of 2025–26).

- *Historical simulation passes, with a lag.* It captures the empirical tail (35 exceptions at 99%, $p = 0.13$) but adapts as slowly as its window: at 95% it collected its surplus of exceptions in March 2020 and again in early 2026.
- *Filtered HS fits best.* Empirical tails rescaled by the EWMA volatility give 135 exceptions at 95% (134 expected, $p = 0.95$) and 33 at 99% ($p = 0.25$), with no sign of clustering at 95%.

At 99%, the independence test gives p between 0.06 and 0.09 for all methods: some clustering remains, but about 30 exceptions are too few to say more. With 2686 test days, the Kupiec test detects a 99% VaR with twice the nominal exception rate in 99% of the cases, against 13% on the course data (Table 6): the long backtest is the one to trust.

Table 6: Power of the Kupiec test (5% significance): course backtest and public-data backtest.

VaR level	true exception rate	132 days	2686 days
95%	7.5%	19%	100%
95%	10.0%	57%	100%
95%	15.0%	97%	100%
99%	1.5%	5%	66%
99%	2.0%	13%	99%
99%	3.0%	36%	100%

7 Main changes with respect to the course notebook

The notebook written for the exercise did not yet answer it. Table 7 lists the corrections.

Table 7: Main changes with respect to the original notebook.

	course notebook	this version
Alignment	returns of silver and EUR/USD paired by row position, not by date (correlation -0.13)	series aligned by date (correlation $+0.48$)
Correlated shocks	Cholesky factor applied to draws already scaled by mean and volatility, volatility applied a second time when revaluing	Cholesky factor of the covariance applied to standard shocks
Risk measure	95th percentile of each factor's return separately; one simulated day revalued	quantile and tail mean of the loss of the position
Units	500 multiplied by a price per troy ounce	$500\,\mathrm{t} = 16.1 \times 10^6$ troy ounces
Methods	Monte Carlo only	five methods, two levels, two horizons, ES
Validation	none	backtest with Kupiec and Christoffersen tests, power analysis, on the course year and on eleven years of public data; tests on synthetic data

8 Conclusions

1. The one-day VaR(95%) of 500 t of silver held in a euro balance sheet was EUR 8.3 m in November 2023, 2.5 % of its value. The positive correlation of silver with the euro is a natural hedge: without it the VaR would be 18 % higher.

2. For a book of silver and copper, the five methods give one-day 95% VaRs between EUR 8.8 and 10.4 m. Model choice matters in the tail: for the ten-day 99% ES the estimates range from EUR 41 to 59 m, and the methods that capture fat tails or changing volatility give the higher values.

3. Diversification between the two commodities lowers the VaR by 27 %.

4. All methods pass the backtest on the course data, but half a year of backtest cannot tell good from mediocre models at the 99% level. On eleven years of public data it can: normal tails underestimate the 99% VaR (45–48 exceptions instead of 27), constant volatility produces clustered exceptions, and filtered historical simulation is the method to use for this book.

References

Philippe Artzner, Freddy Delbaen, Jean-Marc Eber, and David Heath. Coherent measures of risk. *Mathematical Finance*, 9(3):203–228, 1999.

Juan Carlos Chavarría Morales. commodity-inventory-risk: market risk of a commodity inventory. https://github.com/JuanCaChavaMor/commodity-inventory-risk, 2026.

Peter F. Christoffersen. Evaluating interval forecasts. *International Economic Review*, 39(4): 841–862, 1998.

John Hull and Alan White. Incorporating volatility updating into the historical simulation method for value-at-risk. *Journal of Risk*, 1(1):5–19, 1998.

J.P. Morgan/Reuters. Riskmetrics – technical document. Technical report, J.P. Morgan and Reuters, New York, 1996.

Paul H. Kupiec. Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2):73–84, 1995.

Alexander J. McNeil, Rüdiger Frey, and Paul Embrechts. *Quantitative Risk Management: Concepts, Techniques and Tools*. Princeton University Press, revised edition, 2015.

Dirk Tasche. Risk contributions and performance measurement. Working paper, Technische Universität München, 1999.